Zhiping WANG

Ph.D. Student in Photonics, EPFL

wang-zhiping.github.io | 0009-0004-9890-9809
Wang-Zhiping | zhipingwang818@gmail.com
ZhipingWang818 | zhiping.wang@epfl.ch

Education

Doctoral Program in Photonics, École polytechnique fédérale de Lausanne (EPFL) — *Aug. 2026 – Present*
Laboratory of Applied Photonics Devices (LAPD) — *Lausanne, Switzerland*
Ph.D. student supervised by Prof. Christophe Moser
Research focus: computational phase imaging for light-based additive manufacturing.

Master of Science in Physics of Life, University of Basel — *Sep. 2024 – Jun. 2026*
Biozentrum — *Basel, Switzerland*
Biozentrum Master's Fellowship.

Bachelor of Science in Physics, Lanzhou University — *Sep. 2020 – Jul. 2024*
School of Physical Science and Technology — *Lanzhou, China*
Physics (Magnetism Track);

Honors and Awards

EPFL Excellence in Engineering Fellowship	EPFL	*Jun. – Sep. 2025*
Biozentrum Master's Fellowship	University of Basel	*2024 – 2026*
Excellent Bachelor's Thesis	Lanzhou University	*Jun. 2024*
Outstanding Student Scholarship	Lanzhou University	*Sep. 2023 & Sep. 2022*
China Undergraduate Physics Tournament (Northwest Region)	First & Second Prize	*Jul. 2022 & Jun. 2021*

Publications

[1] Zhiping Wang, Tianci Feng, Aiye Wang, Jinghao Xu, and An Pan. "Fusion-based enhancement of multi-exposure Fourier ptychographic microscopy." *Advanced Photonics Nexus* 4, no. 4 (2025): 046001. DOI: 10.1117/1.APN.4.4.046001.

[2] Fannuo Xu†, Zhiping Wang†, Zipei Wu, and An Pan. "Wide-field quantitative phase imaging without slicing via feature-domain Fourier ptychographic microscopy." *Optics & Laser Technology* 187 (2025): 112727. DOI: 10.1016/j.optlastec.2025.112727. (†Equal contribution)

[3] Tianci Feng, Aiye Wang, Zhiping Wang, Yizheng Liao, and An Pan. "Linear-space-variant model for Fourier ptychographic microscopy." *Optics Letters* 49, no. 10 (2024): 2617–2620. DOI: 10.1364/OL.522745.

[4] Fannuo Xu, Zipei Wu, Chao Tan, Yizheng Liao, Zhiping Wang, Keru Chen, and An Pan. "Fourier ptychographic microscopy 10 years on: a review." *Cells* 13, no. 4 (2024): 324. DOI: 10.3390/cells13040324.

[5] Yanqi Chen, Jiurun Chen, Zhiping Wang, Yuting Gao, Yonghong He, Yishi Shi, and An Pan. "Fast full-color pathological imaging using Fourier ptychographic microscopy via closed-form model-based colorization." *Advanced Photonics Nexus* 4, no. 2 (2025): 026001. DOI: 10.1117/1.APN.4.2.026001.

Research Experience

Computational Phase Imaging for Monitoring Light-Based Additive Manufacturing — *Aug. 2026 – Present*
Ph.D. Research, LAPD, EPFL; Supervisor: Prof. Christophe Moser — *Lausanne, Switzerland*

- Developing computational and quantitative phase imaging for in-situ monitoring, with the long-term goal of monitoring TVAM, SHVAM, and holographic fabrication.
- Building a programmable Fourier ptychographic microscopy (FPM) / differential phase contrast (DPC) microscope with sparse multi-angle LED illumination, external camera triggering, and synchronized illumination and acquisition.
- Developing illumination-angle, pupil, aberration, and defocus calibration; investigating vignetting, model mismatch and few-frame imaging for dynamic processes.
- Validating with USAF resolution targets, glass microspheres, and quantitative phase targets.

3D Multi-View Tracking and Behavioral Analysis of *Drosophila* *Oct. 2025 – Jun. 2026*
Master's Thesis, Biozentrum, University of Basel
Advisors: Prof. Anissa Kempf and Dr. Alexandra Medeiros *Basel, Switzerland*

- Built a mirror-based multi-view arena and trained DeepLabCut keypoint models; reconstructed metric 3D trajectories using calibration, triangulation, and outlier filtering, with MuJoCo/PyTorch STAC inverse kinematics for joint-angle analysis.
- Developed locomotion, grooming, immobility, and sleep classifiers; applied hBehaveMAE to unsupervised repertoire discovery and an SVM to proboscis extension ($> 90\%$ accuracy).

3D Reconstruction for TVAM 3D-Printed Samples *Jun. 2025 – Sep. 2025*
EPFL EEE Summer Project; Advisors: Felix Wechsler and Prof. Christophe Moser *Lausanne, Switzerland*

- Developed a multi-slice forward model for 3D FPM using angular-spectrum propagation and implicit neural representation (INR)-based inverse reconstruction in PyTorch; compared INR and voxel representations, improving fidelity and reducing artifacts in simulations.
- Independently assembled an FPM setup with programmable LED illumination, optical alignment, acquisition, and reconstruction software; validated super-resolution using experimental data.

Optimal Filtered Back-Projection for Cryo-Electron Tomography *Nov. 2024 – Feb. 2025*
Master's Project, Biozentrum & Department of Mathematics and Computer Science, University of Basel
Advisors: Dr. Ricardo Righetto and Dr. Valentin Debarnot *Basel, Switzerland*

- Implemented ODL filtered back-projection and PyTorch filter learning with total-variation regularization; improved structural recovery on experimental *Chlamydomonas* data (EMPIAR-11830) compared with conventional filtered back-projection.
  Project report (GitHub).

High-Efficiency Fourier Ptychographic Microscopy *Aug. 2023 – Jul. 2024*
Research Internship, Pioneering Interdiscipline Center, Chinese Academy of Sciences
Supervisor: Dr. An Pan *China*

- Investigated acquisition efficiency and wide-field QPI; identified block-stitching artifacts and expanded the reconstructed field of view from approximately 0.2 to 3.3 mm through simulation, calibration, and programmable LED illumination.
- Developed nonlinear multi-exposure fusion (MEIF) for low-SNR dark-field data and contributed to feature-domain reconstruction; first-author *Advanced Photonics Nexus* and co-first-author *Optics & Laser Technology* publications.
  Selected results: project page 1; project page 2.

Skills

Programming	Python (PyTorch, TensorFlow, OpenCV, NumPy, pandas), MATLAB, Mathematica, C/C++, LaTeX
Computational Imaging / Scientific Computing	FPM, QPI, phase retrieval, inverse reconstruction, image processing, CUDA/GPU, HPC/SLURM
Tools	DeepLabCut, MuJoCo, Autodesk Inventor

Teaching Experience

École polytechnique fédérale de Lausanne (EPFL) *Sep. 2026 – Present*
Teaching Assistant, ME-413 Introduction to Additive Manufacturing *Lausanne, Switzerland*

School of Physical Science and Technology, Lanzhou University *Sep. 2021 – Jan. 2022*
Teaching Assistant, Computational Physics *Lanzhou, China*

- Graded assignments, held office hours, and assisted with computational exercises and course materials.